

Ecole Supérieure en Informatique
ESI-Sidi Bel Abbès

Test Report

Computer Systems and Network Security
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Ex01 : Research Workspace ACL

List of ACLs :

```
(kali㉿kali)-[/home/manel_mostefaoui]
$ sudo setfacl -m d:u:emma:rwX acl1

(kali㉿kali)-[/home/manel_mostefaoui]
$
```

```
(kali㉿kali)-[/home/manel_mostefaoui]
$ sudo setfacl -m m::X acl1

(kali㉿kali)-[/home/manel_mostefaoui]
$ getfacl acl1
# file: acl1
# owner: emma
# group: root
user::rwX
user:emma:rwX          #effective:--X
user:liam:rwX          #effective:--X
user:sophia:r--        #effective:---
user:noah:-w-          #effective:---
group::r-X             #effective:--X
mask::--X
other::r-X
default:user::rwX
default:user:emma:rwX
default:group::r-X
default:mask::rwX
default:other::r-X
```

Ex02: Tcpdumb wireshark

```
(kali㉿kali)-[/home/manel_mostefaoui]
$ sudo tcpdump -i eth0 -w captureA.pcap
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
^C46 packets captured
46 packets received by filter
0 packets dropped by kernel
```

```
(kali@kali) - [~/home/manel_mostefaoui]
$ wireshark captureA.pcap
** (wireshark:423299) 05:30:22.111076 [GUI ECHO] -- virtual const QPalette* Qt6CTPlatformTheme::palette(QPlatformTheme::Palette) d
onst QPlatformTheme::SystemPalette
** (wireshark:423299) 05:30:22.112119 [GUI ECHO] -- virtual const QPalette* Qt6CTPlatformTheme::palette(QPlatformTheme::Palette) d
onst QPlatformTheme::ToolButtonPalette
** (wireshark:423299) 05:30:22.112289 [GUI ECHO] -- virtual const QPalette* Qt6CTPlatformTheme::palette(QPlatformTheme::Palette) d
onst QPlatformTheme::ButtonPalette
** (wireshark:423299) 05:30:22.112908 [GUI ECHO] -- virtual const QPalette* Qt6CTPlatformTheme::palette(QPlatformTheme::Palette) d
onst QPlatformTheme::CheckBoxPalette
```

[illegible]

No.	Time	Source	Destination	Protocol	Length	Info
4	0.001360	192.168.24.128	192.168.24.129	HTTP	144	GET / HTTP/1.1
12	0.358539	192.168.24.129	192.168.24.128	HTTP	71	HTTP/1.1 200 OK (text/html)
21	0.555550	192.168.24.128	192.168.24.129	HTTP	144	GET / HTTP/1.1
25	1.618491	192.168.24.129	192.168.24.128	HTTP	71	HTTP/1.1 200 OK (text/html)
33	2.667124	192.168.24.128	192.168.24.129	HTTP	144	GET / HTTP/1.1
37	2.733865	192.168.24.129	192.168.24.128	HTTP	71	HTTP/1.1 200 OK (text/html)
47	3.607987	192.168.24.128	192.168.24.129	HTTP	144	GET / HTTP/1.1
51	3.648997	192.168.24.129	192.168.24.128	HTTP	71	HTTP/1.1 200 OK (text/html)
59	4.446256	192.168.24.128	192.168.24.129	HTTP	144	GET / HTTP/1.1
62	4.496561	192.168.24.129	192.168.24.128	HTTP	71	HTTP/1.1 200 OK (text/html)
75	1.223298	192.168.24.128	192.168.24.129	HTTP	144	GET / HTTP/1.1
79	5.381845	192.168.24.129	192.168.24.128	HTTP	71	HTTP/1.1 200 OK (text/html)
83	6.141290	192.168.24.128	192.168.24.129	HTTP	144	GET / HTTP/1.1
87	6.168681	192.168.24.129	192.168.24.128	HTTP	71	HTTP/1.1 200 OK (text/html)
95	7.185726	192.168.24.128	192.168.24.129	HTTP	144	GET / HTTP/1.1
99	7.236594	192.168.24.129	192.168.24.128	HTTP	71	HTTP/1.1 200 OK (text/html)
111	8.388550	192.168.24.128	192.168.24.129	HTTP	144	GET / HTTP/1.1

Frame 4: 144 bytes on wire (1152 bits), 144 bytes captured (1152 bits)

Ethernet II, Src: VMware_53:d5:25 (00:0c:29:53:d5:25), Dst: VMware_26:b8:9c (00:0c:29:26:b8:9c)

Internet Protocol Version 4, Src: 192.168.24.128, Dst: 192.168.24.129

Transmission Control Protocol, Src Port: 59280, Dst Port: 80, Seq: 1, Ack: 1, Len: 78

Hypertext Transfer Protocol

0000 00 0c 29 26 b8 9c 00 0c 29 53 d5 25 08 00 45 00 --) & % 5 %

0010 00 82 17 ca 40 08 09 46 70 5a c0 a8 18 80 c0 a8 --) @ @ pZ %

0020 18 01 e7 9a 00 56 7b f9 34 77 51 1b ac 74 80 18 --) P 4wQ t

0030 00 00 b2 05 00 00 01 01 80 0a e7 67 f4 80 00 36 --) 4

0040 1d 21 4f 45 54 20 2f 2d 48 54 50 2f 31 3e 21 --) IGET / HTTP/1.1

0050 00 0a 48 45 73 74 3a 20 31 39 32 32 31 36 38 2e --) Host: 192.168.24.129

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.24.128	192.168.24.129	ICMP	74	59280 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3890/38310 TSecr=0 WS=128
2	0.000095	192.168.24.129	192.168.24.128	ICMP	74	80 → 59280 [SYN, ACK] Seq=0 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=3546436 TSecr=3890/38310 WS=64
3	0.001044	192.168.24.128	192.168.24.129	ICMP	66	59280 → 80 [ACK] Seq=1 Ack=1 Win=64260 Len=0 TSval=3890/38317 TSecr=3546436
4	0.001100	192.168.24.129	192.168.24.128	ICMP	66	80 → 59280 [ACK] Seq=1 Ack=1 Win=5824 Len=0 TSval=3546461 TSecr=3890/38317
5	0.002188	192.168.24.129	192.168.24.128	ICMP	66	80 → 59280 [PSH, ACK] Seq=1 Ack=1 Win=5824 Len=507 TSval=3546436 TSecr=3890/38317 [TCP PDU reassembled in 12]
6	0.352077	192.168.24.129	192.168.24.128	ICMP	499	80 → 59280 [ACK] Seq=398 Ack=79 Win=5824 Len=24 TSval=3890/38316 TSecr=3890/38317 [TCP PDU reassembled in 12]
7	0.352076	192.168.24.128	192.168.24.129	ICMP	139	80 → 59280 [PSH, ACK] Seq=1022 Ack=79 Win=5824 Len=24 TSval=3546436 TSecr=3890/38317 [TCP PDU reassembled in 12]
8	0.352174	192.168.24.128	192.168.24.129	ICMP	66	59280 → 80 [ACK] Seq=9 Ack=508 Win=63744 Len=0 TSval=3890/38608 TSecr=3546436
9	0.352275	192.168.24.128	192.168.24.129	ICMP	66	59280 → 80 [ACK] Seq=9 Ack=1022 Win=63600 Len=0 TSval=3890/38608 TSecr=3546436
10	0.352361	192.168.24.129	192.168.24.128	ICMP	66	59280 → 80 [ACK] Seq=9 Ack=1095 Win=63600 Len=0 TSval=3890/38608 TSecr=3546436
11	0.350859	192.168.24.129	192.168.24.128	HTTP	71	HTTP/1.1 200 OK. [text/html]
12	0.350862	192.168.24.128	192.168.24.129	ICMP	66	59280 → 80 [ACK] Seq=9 Ack=1100 Win=63600 Len=0 TSval=3890/38674 TSecr=3546437
13	0.350954	192.168.24.128	192.168.24.129	ICMP	66	59280 → 80 [FIN, ACK] Seq=9 Ack=1100 Win=63600 Len=0 TSval=3890/38675 TSecr=3546437
14	0.404006	192.168.24.129	192.168.24.128	ICMP	66	80 → 59280 [ACK] Seq=1100 Ack=80 Win=5824 Len=0 TSval=3546436 TSecr=3890/38675
15	0.404278	192.168.24.129	192.168.24.128	ICMP	66	80 → 59280 [FIN, ACK] Seq=1100 Ack=80 Win=5824 Len=0 TSval=3546442 TSecr=3890/38675
16	0.404361	192.168.24.128	192.168.24.129	ICMP	66	59280 → 80 [ACK] Seq=80 Ack=1101 Win=63600 Len=0 TSval=3890/38720 TSecr=3546442

Ex03: NMAP basic host discovery

```
(kali㉿kali)-[/home/manel_mostefaoui]
$ nmap -p 1-100 192.168.24.129
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-10 05:45 EST
Stats: 0:00:11 elapsed; 0 hosts completed (0 up), 1 undergoing ARP Ping Scan
Parallel DNS resolution of 1 host. Timing: About 0.00% done
Nmap scan report for 192.168.24.129
Host is up (0.0026s latency).
Not shown: 94 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
MAC Address: 00:0C:29:26:B8:9C (VMware)

Nmap done: 1 IP address (1 host up) scanned in 13.29 seconds
```

```
(kali㉿kali)-[/home/manel_mostefaoui]
$ nmap -sC -sV 192.168.24.129
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-10 05:53 EST
Stats: 0:00:06 elapsed; 0 hosts completed (0 up), 1 undergoing ARP Ping Scan
Parallel DNS resolution of 1 host. Timing: About 0.00% done
Stats: 0:01:04 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 86.96% done; ETC: 05:54 (0:00:08 remaining)
Stats: 0:01:07 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 86.96% done; ETC: 05:54 (0:00:08 remaining)
Stats: 0:01:27 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 86.96% done; ETC: 05:54 (0:00:11 remaining)
Stats: 0:03:42 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 99.88% done; ETC: 05:56 (0:00:00 remaining)
Stats: 0:04:22 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 94.68% done; ETC: 05:57 (0:00:02 remaining)
Nmap scan report for 192.168.24.129
Host is up (0.0019s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
| ftp-syst:
|   STAT:
| FTP server status:
|   Connected to 192.168.24.128
|   Logged in as ftp
|   TYPE: ASCII
|   No session bandwidth limit
```

Services and version that are running :

```
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ftp-syst:
|_STAT:
|_FTP server status:
|_    Connected to 192.168.24.128
|_    Logged in as ftp
|_    TYPE: ASCII
|_    No session bandwidth limit
|_    Session timeout in seconds is 300
|_    Control connection is plain text
|_    Data connections will be plain text
|_    vsFTPd 2.3.4 - secure, fast, stable
|_End of status
```

```
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ssh-hostkey:
|_ 1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_ 2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDST
ATUSCODES, 8BITMIME, DSN
53/tcp    open  domain       ISC BIND 9.4.2
|_dns-nsid:
|_bind.version: 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_http-title: Metasploitable2 - Linux
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
111/tcp   open  rpcbind      2 (RPC #100000)
|_rpcinfo:
|_  program version    port/proto  service
|_  100000  2                111/tcp    rpcbind
|_  100000  2                111/udp    rpcbind
|_  100003  2,3,4           2049/tcp   nfs
|_  100003  2,3,4           2049/udp   nfs
|_  100005  1,2,3           38224/tcp  mountd
|_  100005  1,2,3           42032/udp  mountd
|_  100021  1,3,4           45549/udp  nlockmgr
|_  100021  1,3,4           52186/tcp  nlockmgr
|_  100024  1                48067/udp  status
|_  100024  1                50899/tcp  status
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rshcd
513/tcp   open  login?
514/tcp   open  shell        Netkit rshd
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
```

Ex04: Basic UFW

```
(kali㉿kali)-[/home/manel_mostefaoui]
$ sudo ufw enable
[sudo] password for kali:
Firewall is active and enabled on system startup
```

```
(kali㉿kali)-[/home/manel_mostefaoui]
$ sudo ufw allow 80/tcp

Rule added
Rule added (v6)

(kali㉿kali)-[/home/manel_mostefaoui]
$ sudo ufw allow 22/tcp
Skipping adding existing rule
Skipping adding existing rule (v6)
```

```
(kali㉿kali)-[/home/manel_mostefaoui]
$ sudo ufw status numbered
Status: active
```

	To	Action	From
	--	-----	-----
[1]	22/tcp	ALLOW IN	Anywhere
[2]	80/tcp	ALLOW IN	Anywhere
[3]	22/tcp (v6)	ALLOW IN	Anywhere (v6)
[4]	80/tcp (v6)	ALLOW IN	Anywhere (v6)